the following sections – isn't about consciousness. Rather, the book argues that the emergence of computers capable of pursuing goals and making decisions by themselves changes the fundamental structure of our information network.

LINKS IN THE CHAIN

Prior to the rise of computers, humans were indispensable links in every chain of information networks like churches and states. Some chains were composed only of humans. Muhammad could tell Fatima something, then Fatima told Ali, Ali told Hasan and Hasan told Hussain. This was a human-to-human chain. Other chains included documents, too. Muhammad could write something down, Ali could later read the document, interpret it and write his interpretation in a new document, which more people could read. This was a human-to-document chain.

But it was utterly impossible to create a document-to-document chain. A text written by Muhammad could not produce a new text without the help of at least one human intermediary. The Quran couldn't write the Hadith, the Old Testament couldn't compile the Mishnah and the US Constitution couldn't compose the Bill of Rights. No paper document has ever produced by itself another paper document, let alone distributed it. The path from one document to another must always pass through the brain of a human.

In contrast, computer-to-computer chains can now function without humans in the loop. For example, one computer might generate a fake news story and post it on a social media feed. A second computer might identify this as fake news and not just delete it but also warn other computers to block it. Meanwhile, a third computer analyzing this activity might deduce that this indicates the beginning of a political crisis, and immediately sell risky stocks and buy safer government bonds. Other computers monitoring financial transactions may react by selling more stocks, triggering a financial downturn.[29] All this could happen within seconds, before any human can notice and decipher what all these computers are doing.

Another way to understand the difference between computers and all previous technologies is that computers are fully fledged members of the information network, whereas clay tablets, printing presses and radio sets are merely connections between members. Members are active agents that